• Regresión Lineal (Simple)

DATOS

X (features), y (target)

Modelo

$\hat{y} = \beta_0 + \beta_1 X$

$\hat{y}$: predicción

β_0, β_1: Parámetros de ajuste

Error (Función de Pérdida)

$$\mathcal{L}(\beta_0, \beta_1) = \frac{1}{n} \sum_{i=1}^{n} (\hat{y}_i - y_i)^2$$

$\varepsilon_i = \hat{y}_i + y_i$ residuos

→ optimización → $\min(\mathcal{L})$

→ Solución analítica

$$\beta_1 = \frac{\sigma_{xy}}{\sigma_x^2} \; ; \; \beta_0 = \bar{y} - \beta_1 \bar{x}$$

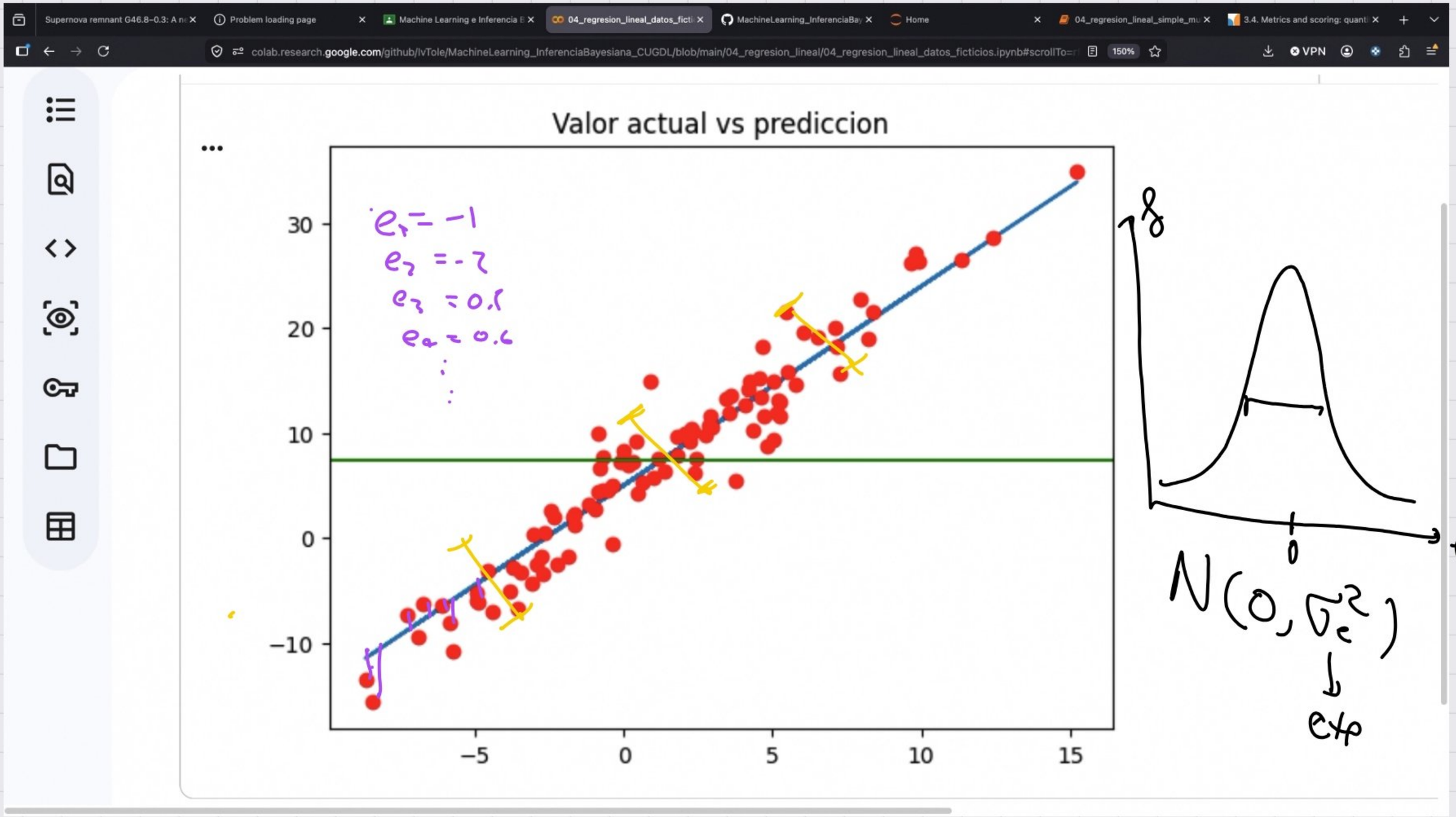
"Aprendizaje"

Error Mínimo

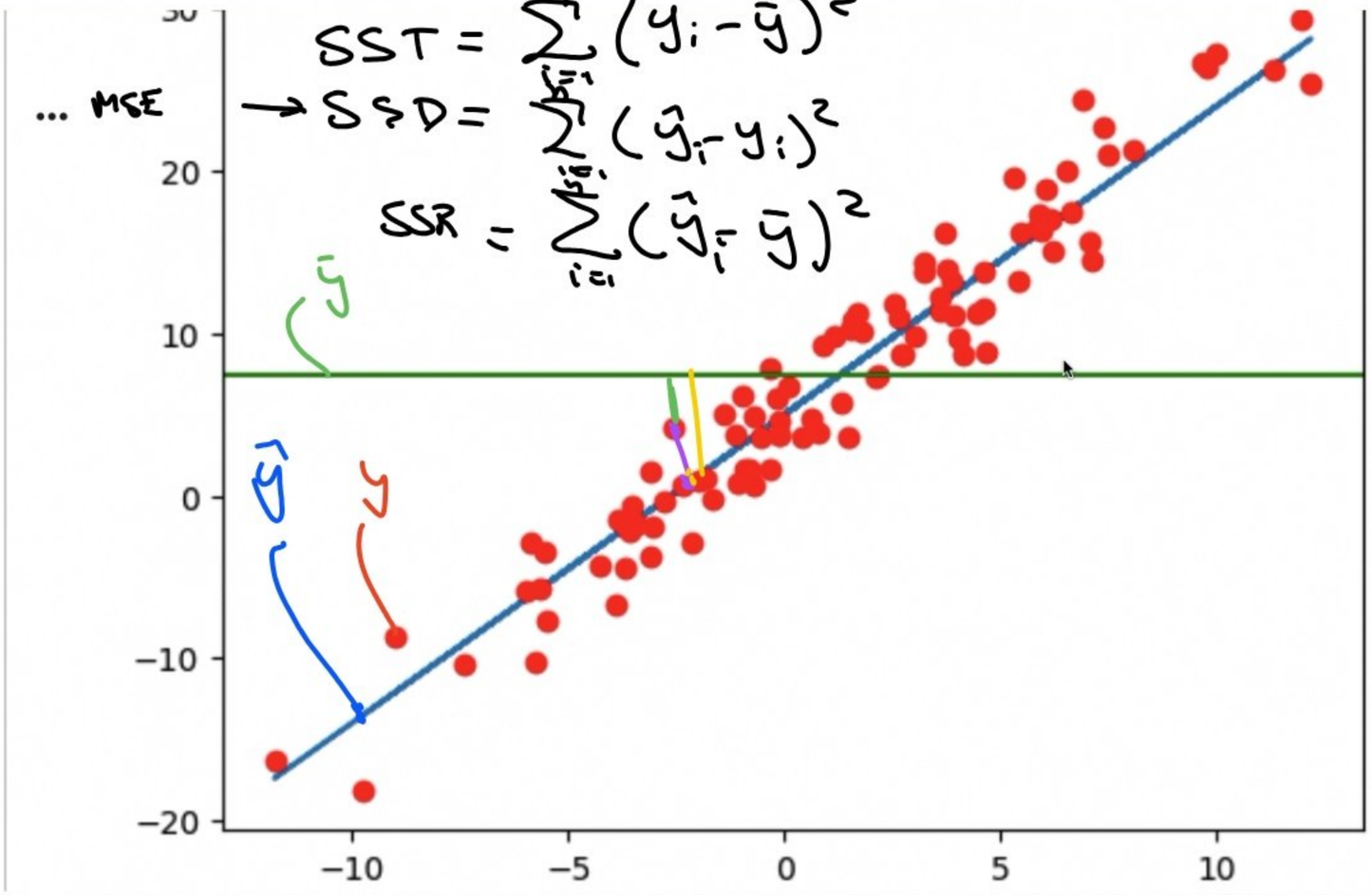
Parámetros óptimos

→ La mejor recta ajustada

$$\hat{y} = \beta_0 + \beta_1 x + \underbrace{\varepsilon}_{\text{error irreducible}}$$


Valor actual vs prediccion
30
20
10
0
−10
−5
0
5
10
15
$e_1 = -1$
$e_2 = -2$
$e_3 = 0.1$
$e_4 = 0.6$
$N(0, \sigma_\epsilon^2)$
exp

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3.4. Metrics and scoring: quanti
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Valor actual vs prediccion

$R^2 \in [0, 1]$

$$SST = \sum_{i=1}^{n} (y_i - \bar{y})^2$$

$$SSD = \sum_{i=1}^{n} (y_i - \hat{y}_i(x_i))^2$$

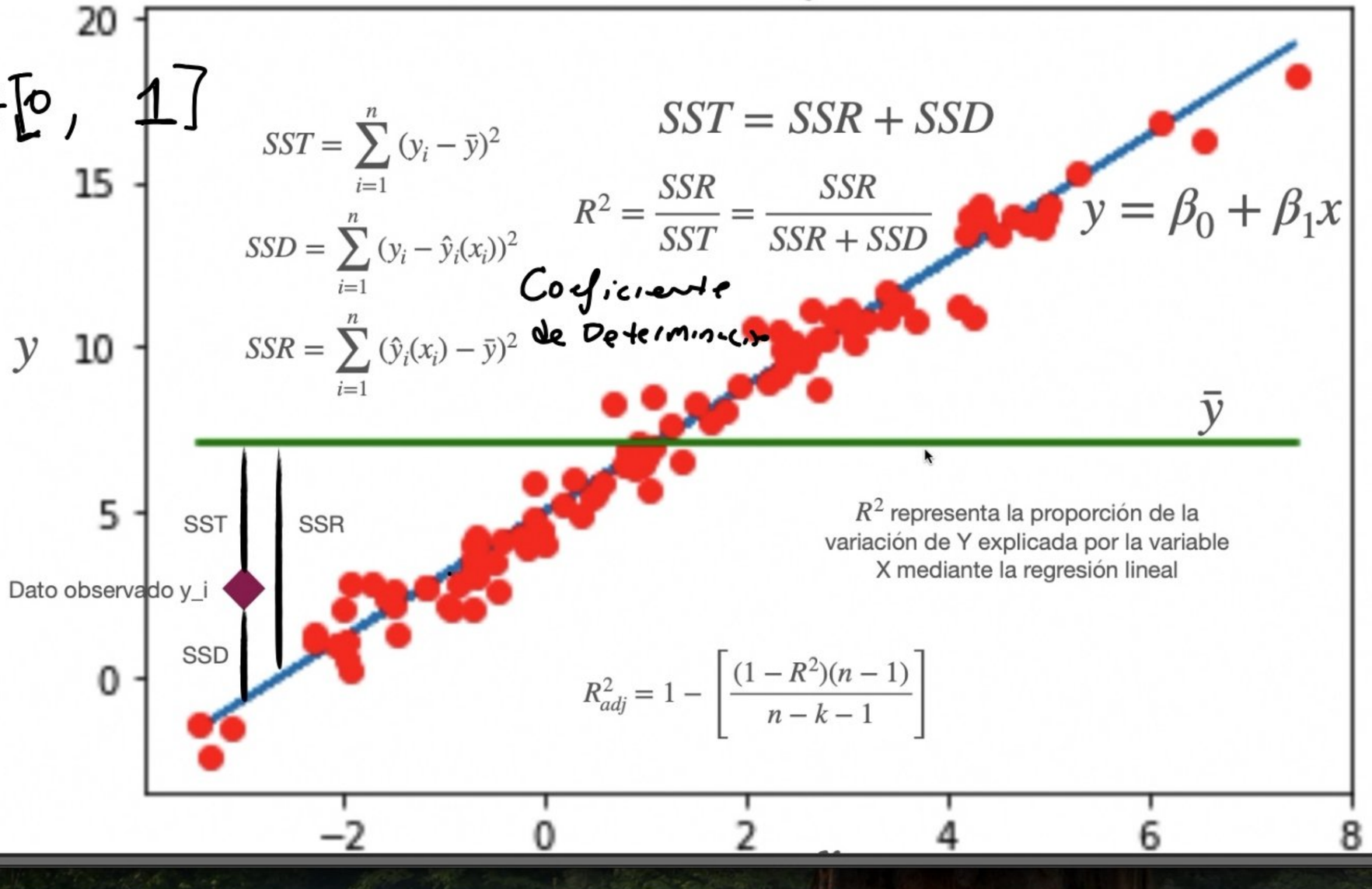
$$SSR = \sum_{i=1}^{n} (\hat{y}_i(x_i) - \bar{y})^2$$

$$SST = SSR + SSD$$

$$R^2 = \frac{SSR}{SST} = \frac{SSR}{SSR + SSD}$$

$$y = \beta_0 + \beta_1 x$$

Coeficiente de Determinación



R^2 representa la proporción de la variación de Y explicada por la variable X mediante la regresión lineal

$$R^2_{adj} = 1 - \left[\frac{(1 - R^2)(n - 1)}{n - k - 1} \right]$$

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colab.research.google.com/github/IvTole/MachineLearning_InferenciaBayesiana_CUGDL/blob/main/04_regresion_lineal/04_regresion_lineal_datos_ficticios.ipynb#scrollTo=2

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04_regresion_lineal_datos_ficticios.ipynb | Save in GitHub to keep changes | Share

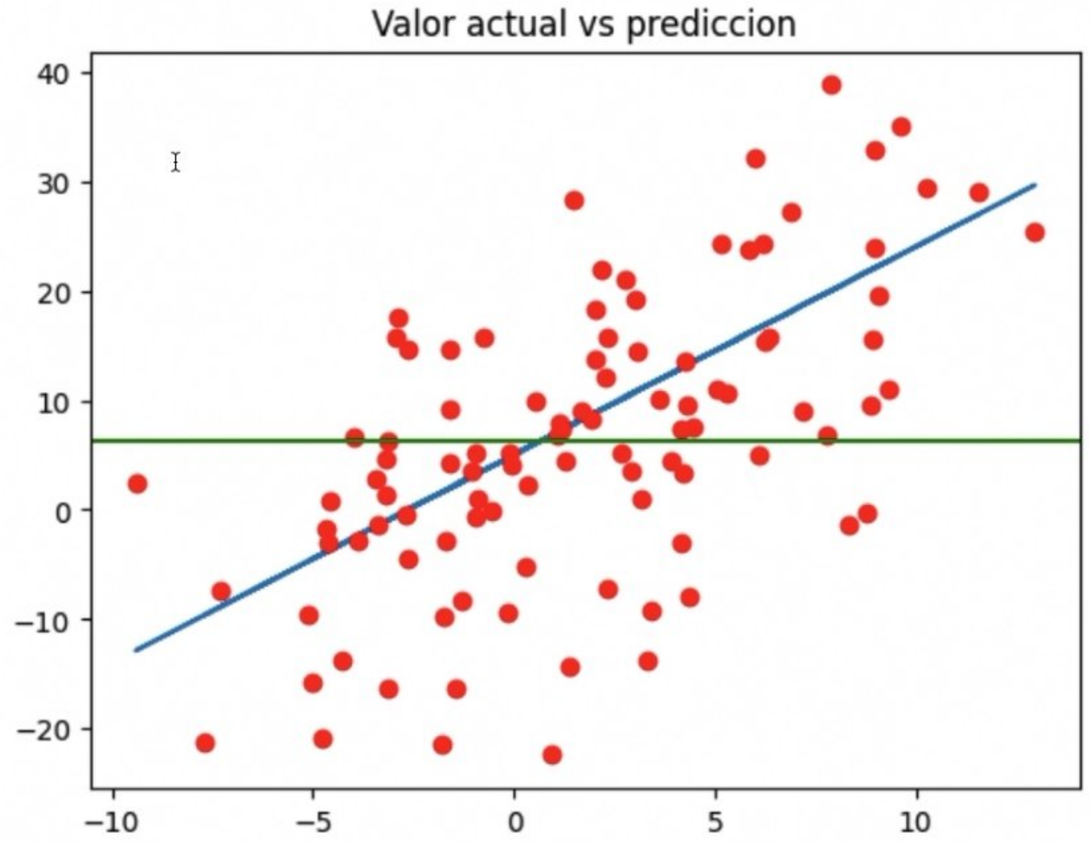
PRO

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```
plt.title("Valor actual vs prediccion")
plt.show()
```

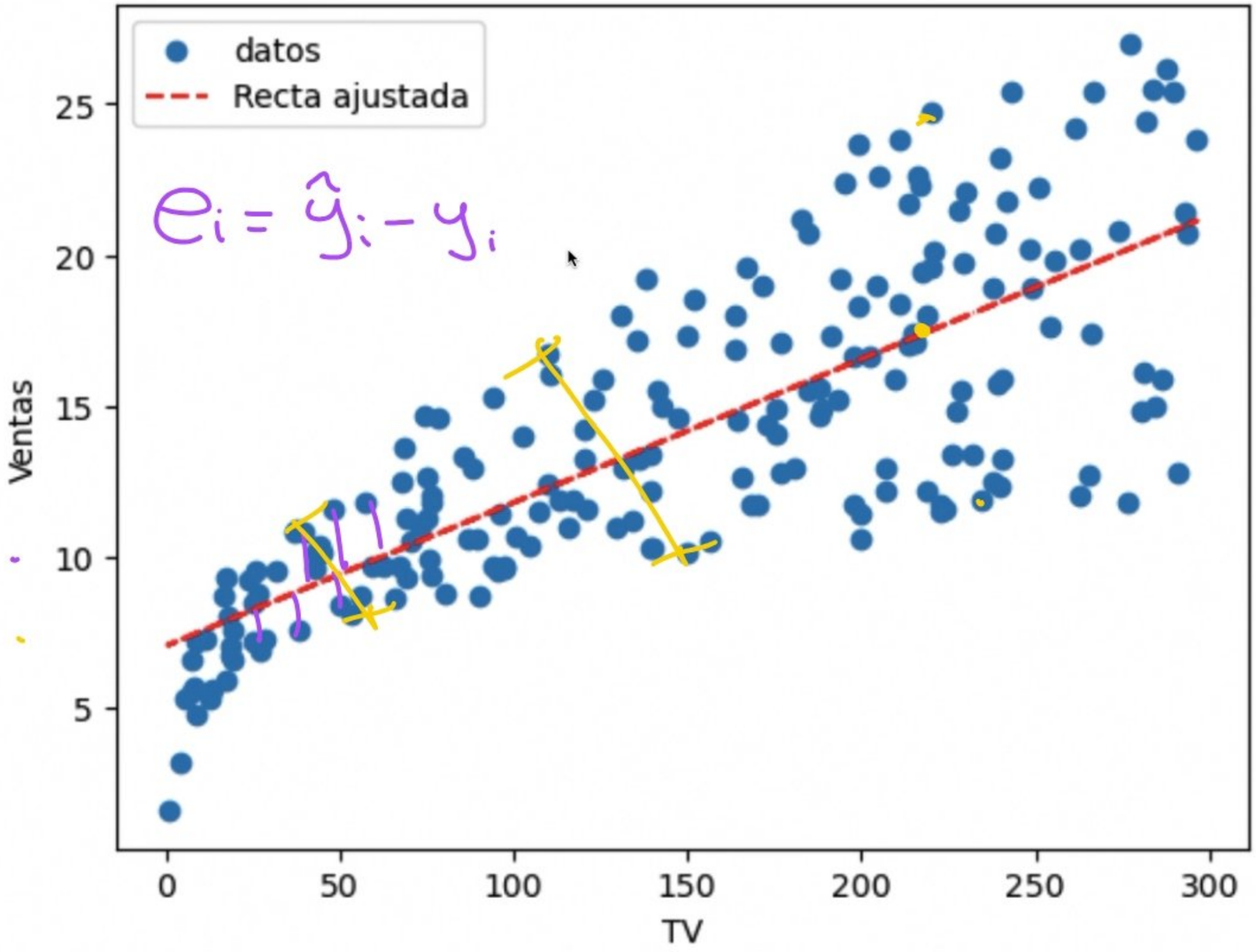


R^2

→ proporción (porcentaje) de la varianza de mis datos que es explicada por mi modelo lineal

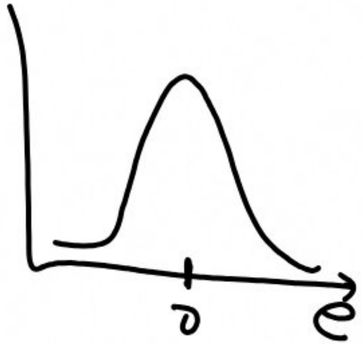
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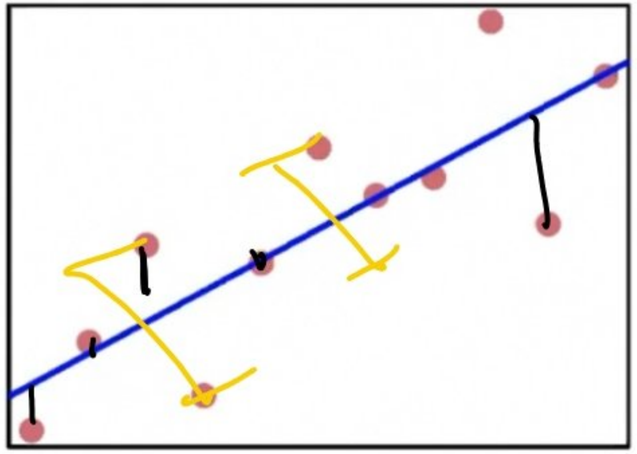
```
Beta0:  7.033
Beta1:  0.048
```



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Visualización Inspector Zoom Compartir Resaltado Girar Marcado Rellenar formulario Buscar Buscar

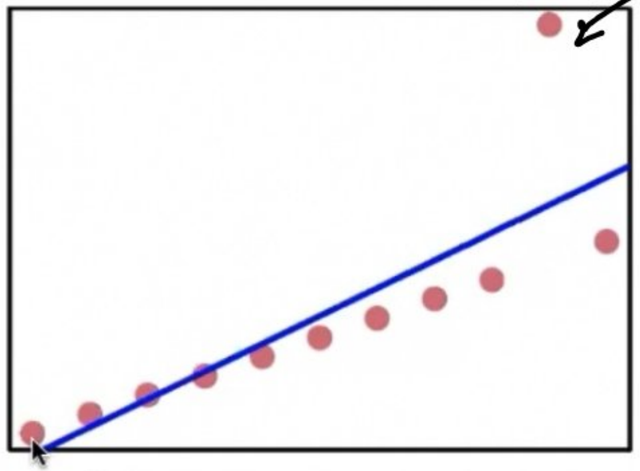
jemplos de algunos modelos ajustados

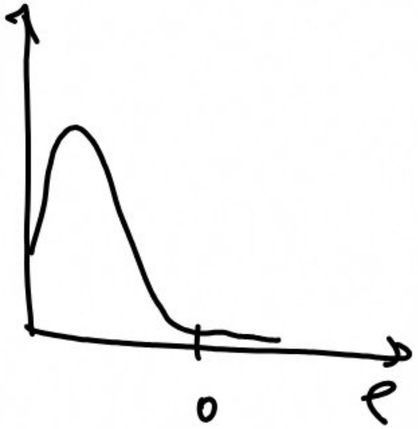




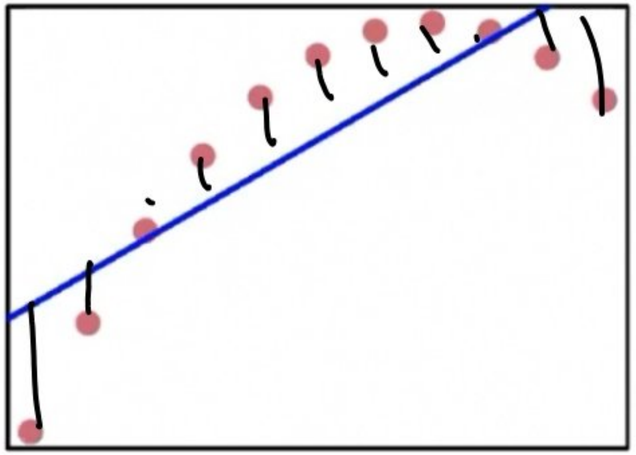
Anscombe's Quartet (1973)

DATO ATIPICO

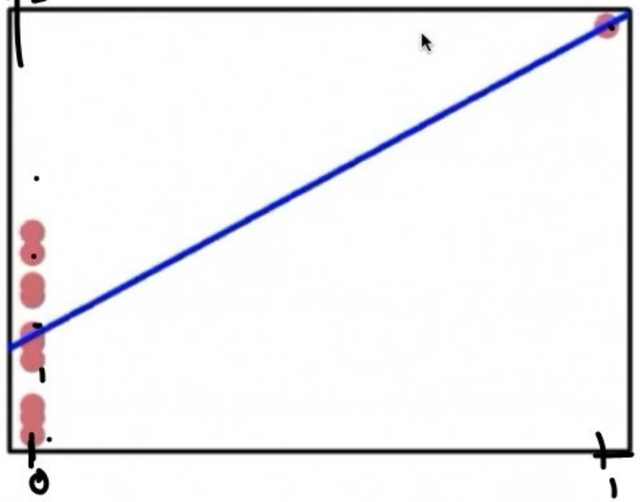


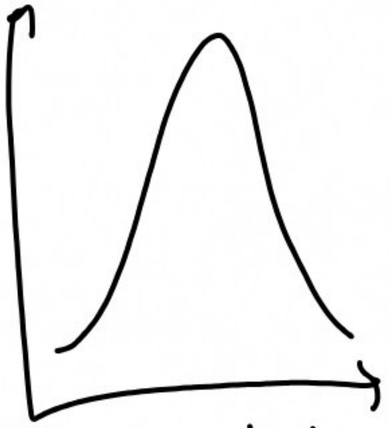






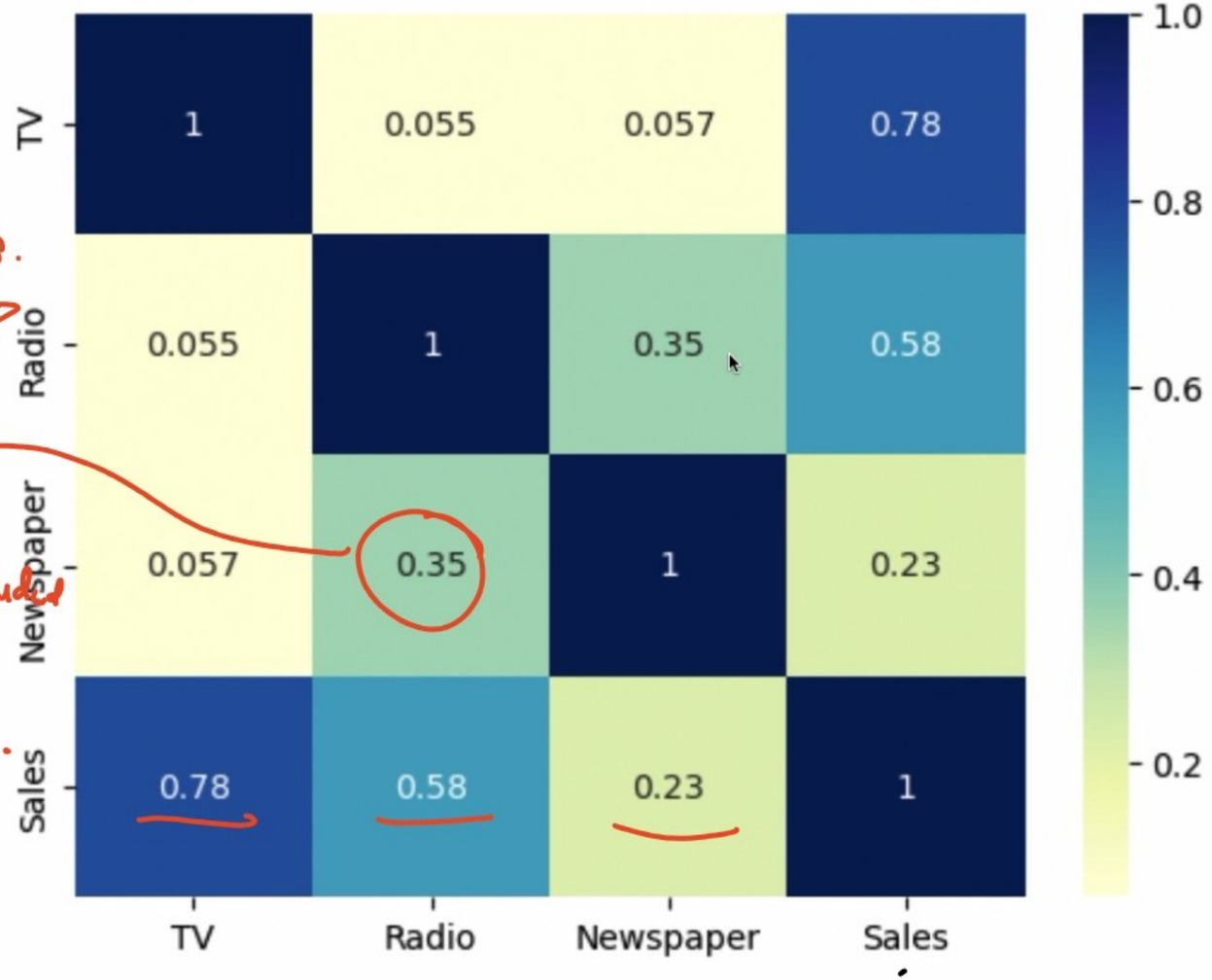
y Scale





0 - Hombre
1 - Mujer

```
plt.show()
```



$\rho_{RADIO,NEWSP.}$

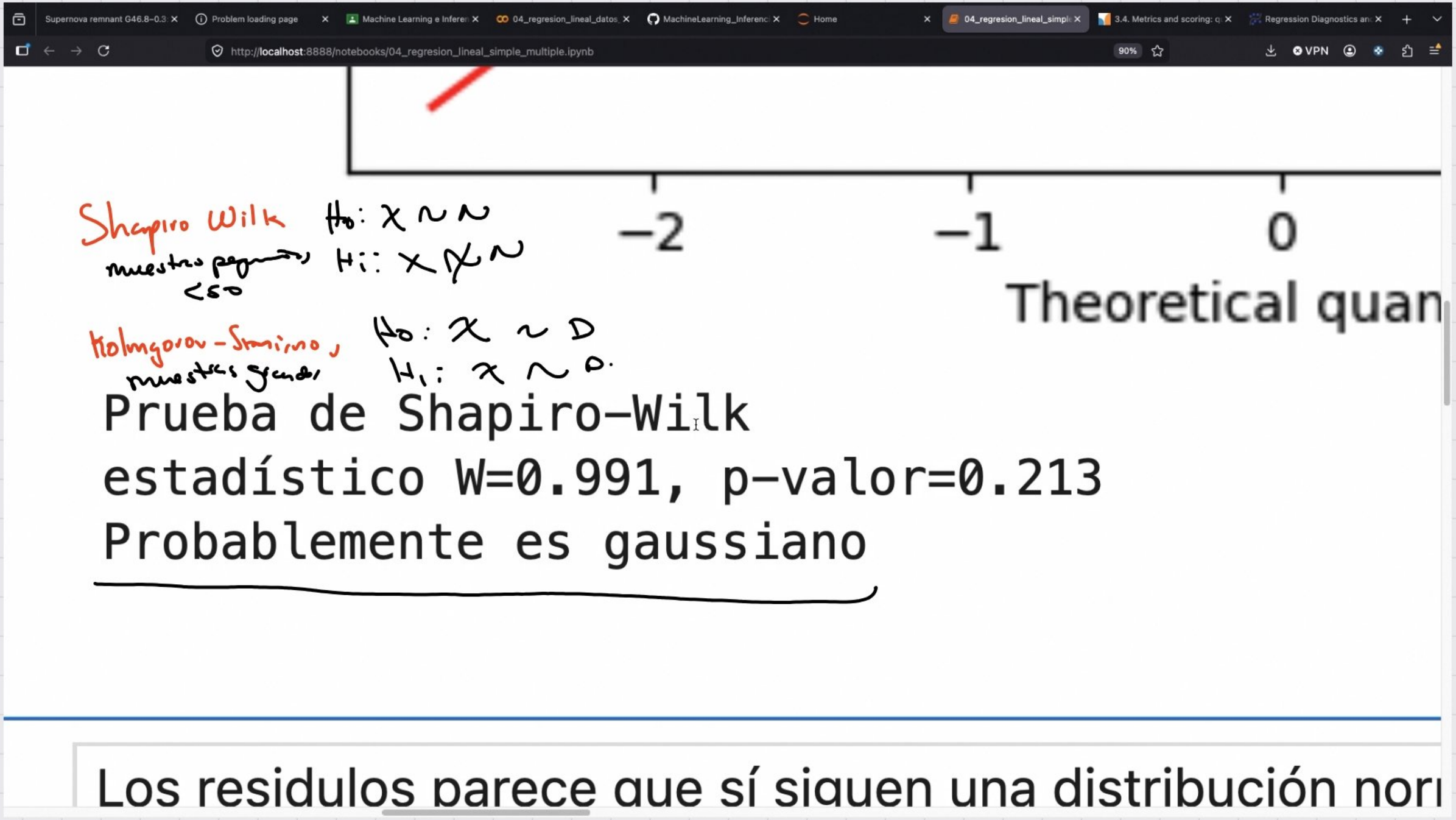
Multicolinealidad

$\rho_{x,y} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$

$\neq$

R^2

De esta matriz de correlación podemos observar que el factor de correlación de Pearson entre la variable `TV` y `Sales` es de 0


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3.4. Metrics and scoring: q
Regression Diagnostics an
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Ho: X ~ N
Hi: X ≁ N
Kolmogorov-Smirnov
muestras grandes
Ho: X ~ D
H1: X ~ D.
−2
−1
0
Theoretical quan
Prueba de Shapiro-Wilk
estadístico W=0.991, p-valor=0.213
Probablemente es gaussiano
Los residulos parece que sí siguen una distribución nor

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[129]:

OLS Regression Results

Dep. Variable:	Sales	**R-squared:**	0.612
Model:	OLS	**Adj. R-squared:**	0.610
Method:	Least Squares	**F-statistic:**	312.1
Date:	Wed, 09 Sep 2026	**Prob (F-statistic):**	1.47e-42
Time:	10:16:22	**Log-Likelihood:**	-519.05
No. Observations:	200	**AIC:**	1042.
Df Residuals:	198	**BIC:**	1049.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	7.0326	0.458	15.360	0.000	6.130	7.935
TV	0.0475	0.003	17.668	0.000	0.042	0.053

Omnibus:	0.531	**Durbin-Watson:**	1.935
Prob(Omnibus):	0.767	**Jarque-Bera (JB):**	0.669

Regresión

> $R^2 \rightarrow \frac{SSR}{SST}$ ⟵ $(y, \hat{y})$

> Residuos $\rightarrow N(0, \sigma^2)$

> Metricas de Evaluación

MSE $\frac{1}{n}\sum (\hat{y}_i - y_i)^2$ (pred, dato)

MAE $\frac{1}{n}\sum |\hat{y}_i - y_i|$

RMSE $\sqrt{\frac{1}{n}\sum (\hat{y}_i - y_i)}$

